Facultad de Estudios Superiores
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The relationship between country size and income per Capita

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Contents

1 Summary

This model examines the relationship between GDP per capita and three variables of development like polpulation size,urbanization and life expectancy using data only from 2019. We use a multiple linear regression model to identify how variations in life expectancy and urban infrastructure explain the wealth gaps between nations.

2019 was selected as the central benchmark because it represents the most stable state of the global economy before COVID-19 pandemic. By focusing in this type of analysis, the objective is not measure growth over the time,, but to understand the structural deifferences that separate developed economies from developing nations in the contemporary era.

1.1 Data Acquisition and Pre processing

The data used in this model were sourced from international repositories consolidating economic and social indicators from official institutions like :

(a) GDP per capita: World bank(via Our World in Data)

(b) Total Population : United Nations Data (via Our World in Data)

(c) Share of Urban Population : (via Our World in Data)

(d) Life Expectancy : (via Our World in Data)

2 Data Wrangling and Cleaning Process (EDA)

First of all we use dynamic CSV files to ensure data traceability and reproducibility.Then we download the data of each of the four variables filtering by the year 2019, then we join the four dataframes with a merge to have a single dataframe with the four variables.

	Entity	Code	Year	GDP per capita	World region according to OWID	Population	Urban population (% of total population)	Life expectancy
0	Afghanistan	AFG	2019	2927.245	Asia	37856125	25.143726	62.9411
1	Albania	ALB	2019	16761.193	Europe	2885011	56.763910	79.4669
2	Algeria	DZA	2019	15199.199	Africa	43294551	72.482450	75.6818
3	Andorra	AND	2019	63215.900	Europe	76492	88.539180	84.0980
4	Angola	AGO	2019	9648.011	Africa	32375633	66.663780	63.0511

In this firts picture are the first five rows of the dataset as a summary.

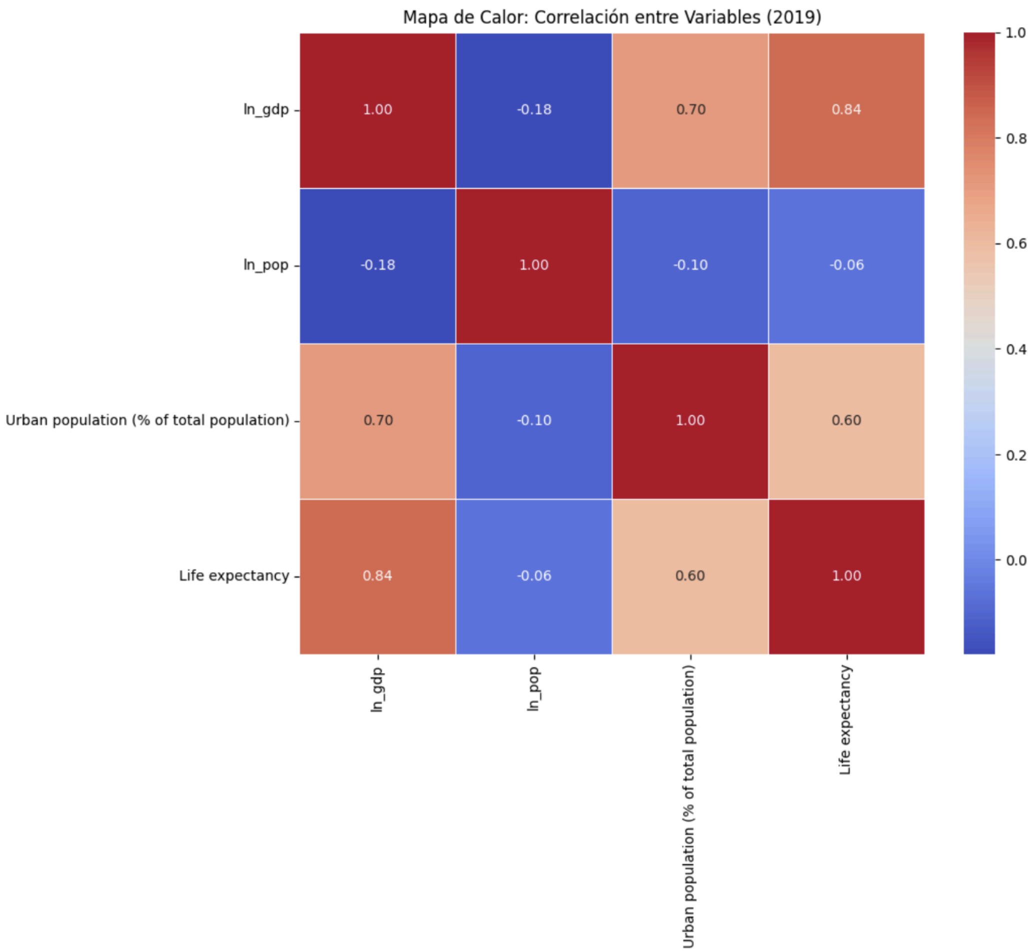
```
<class 'pandas.core.frame.DataFrame'>
Index: 199 entries, 0 to 199
Data columns (total 8 columns):
 #   Column                                    Non-Null Count  Dtype
---  ------                                    --------------  -----
 0   Entity                                    199 non-null    object
 1   Code                                      199 non-null    object
 2   Year                                      199 non-null    int64
 3   GDP per capita                            199 non-null    float64
 4   World region according to OWID            199 non-null    object
 5   Population                                199 non-null    int64
 6   Urban population (% of total population)  199 non-null    float64
 7   Life expectancy                           199 non-null    float64
dtypes: float64(3), int64(2), object(3)
memory usage: 14.0+ KB
```

This sceond image displays the columns in our dataset, showing their data types and missing values were removed to mantein statical integrity.

	Year	GDP per capita	Population	Urban population (% of total population)	Life expectancy
count	199.0	199.000000	1.990000e+02	199.000000	199.000000
mean	2019.0	26305.484065	3.853559e+07	61.255540	72.639930
std	0.0	26223.851145	1.460296e+08	23.126610	7.941055
min	2019.0	1057.950100	1.060300e+04	13.927827	31.530200
25%	2019.0	6337.483350	1.313884e+06	42.591521	66.868800
50%	2019.0	16761.193000	6.966155e+06	62.709915	73.668100
75%	2019.0	38237.003500	2.684955e+07	79.606490	78.260550
max	2019.0	133453.890000	1.423520e+09	100.000000	85.263200

This third image displays the descriptive statistics for the data, including the count, mean, standard deviation, minimum, maximum, and quartiles.
And to correct the right-skewness of the GDP per capita and Population variables, a natural logarithmic transformation was applied. This allows for the interpretation of coefficients as elasticities and ensures the model meets the normality assumptions required for linear regression.

2.1 EDA Graphs



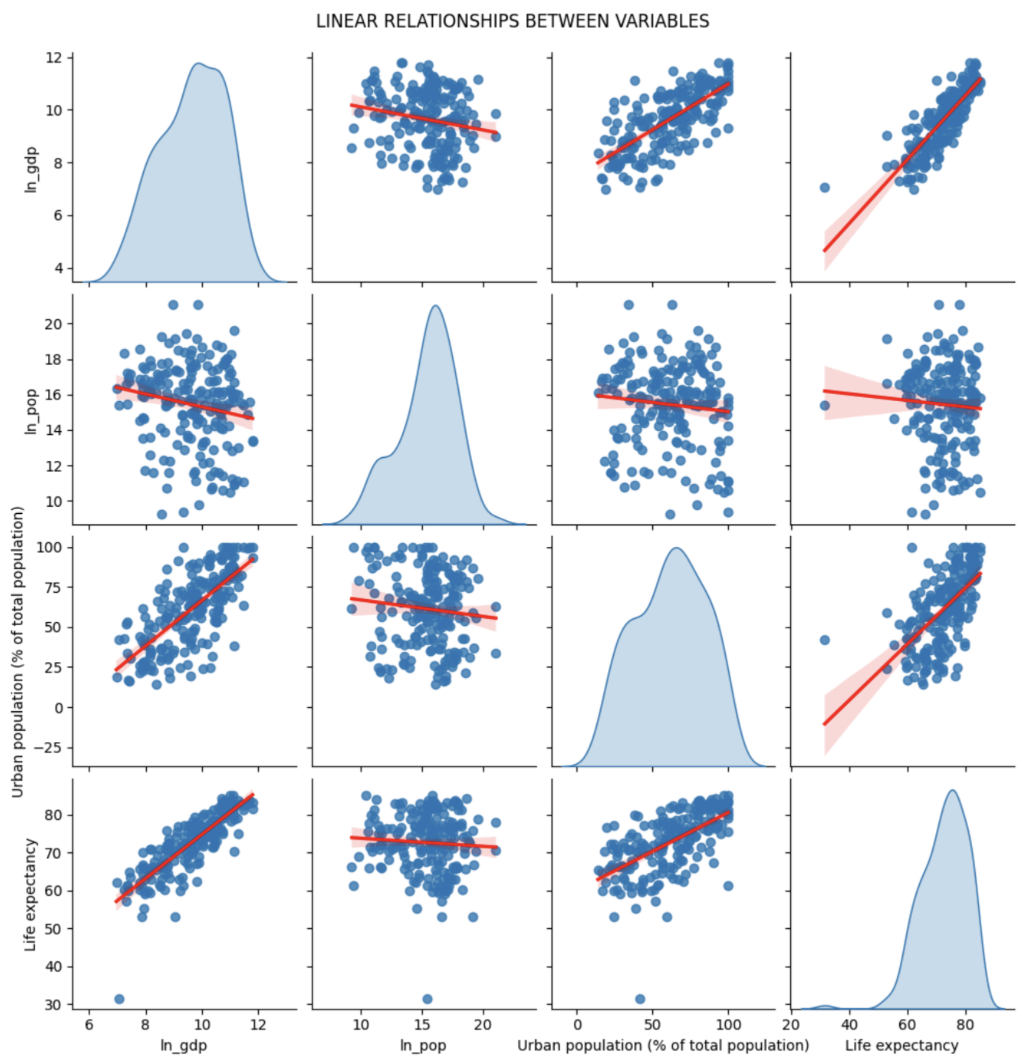
The correlation matrix provides an overview of the relationships between the variables used in the analysis.

GDP per capita (in logarithms) shows a strong positive correlation with life expectancy (0.84) and a high positive correlation with urban population (0.70). This suggests that countries with higher levels of human development and urbanization tend to have higher income per capita.

In contrast, GDP per capita exhibits a weak negative correlation with population (-0.18), indicating that larger countries tend to have slightly lower income per capita, although the relationship is not strong.

Additionally, urban population and life expectancy are positively correlated (0.60), which may reflect that more developed countries tend to be both more urbanized and have better health conditions.

Overall, the correlations are consistent with economic intuition and support the inclusion of these variables in the regression model.



This scatter plots illustrate the pairwise relationships between GDP per capita (in logarithms) and the explanatory variables.

A clear positive relationship is observed between GDP per capita and both urban population and life expectancy, indicating that higher levels of urbanization and better health conditions are associated with higher income levels.

In contrast, the relationship between GDP per capita and population appears to be slightly negative and less pronounced, suggesting a weaker association.

Overall, the patterns observed in the scatter plots appear approximately linear, which supports the use of a linear regression model for the analysis.

3 Model Specification

A Multiple Linear Regression model was constructed. and now, we define our dependent and independent variables.

- **Dependent Variable:**

  GDP per capita: Measures de average income by person in each country, and used as a proxy for economic prosperty.

- **Independent Variables:**

  Population: Represents de size of the country, and used to capture scale effects and demographic structure.

  Urban population: Measures de percentage of people living in urban areas, and proxy for urbanization and economic concentration.

  Life expectancy: Measures de average expected lifespan, and proxy for human development and health conditions.

So,the coefficients of the model are interpreted as follows:

1. β_1 : Represents the elasticity of GDP per capita with respect to population.
2. β_2 : Measures the change in GDP per capita associated with a one percentage point increase in urban population.
3. β_3 : Captures de effect of an additional year of life expectancy on income per capita.

As we said the logarithmic transformation es applied to GDP per capita and Population in order to reduce skewness, stabilize variance, and allow for elasticity interpretation of the coefficients.

The validity of the estimators relies on several assumptions, including linearity, independence of errors, homoscedasticity, and absence of perfect multicollinearity, so these assumptions are evaluated by tests.

4 Evaluating the Model Results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 ln_gdp   R-squared:                       0.778
Model:                            OLS   Adj. R-squared:                  0.775
Method:                 Least Squares   F-statistic:                     228.2
Date:                Thu, 26 Feb 2026   Prob (F-statistic):           1.59e-63
Time:                        04:42:34   Log-Likelihood:                -158.52
No. Observations:                 199   AIC:                             325.0
Df Residuals:                     195   BIC:                             338.2
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================================================
                                              coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------------------------------
const                                       2.7042      0.464      5.824      0.000       1.789       3.620
ln_pop                                     -0.0519      0.017     -3.124      0.002      -0.085      -0.019
Urban population (% of total population)    0.0149      0.002      7.155      0.000       0.011       0.019
Life expectancy                             0.0937      0.006     15.466      0.000       0.082       0.106
==============================================================================
Omnibus:                        0.259   Durbin-Watson:                   2.019
Prob(Omnibus):                  0.879   Jarque-Bera (JB):                0.072
Skew:                          -0.006   Prob(JB):                        0.965
Kurtosis:                       3.092   Cond. No.                     1.19e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.19e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

The regression results indicate that all explanatory variables are statistically significant at conventional levels.

The coefficient on population is negative (-0.0519), suggesting that, within the sample, countries with larger populations tend to have slightly lower levels of income per capita, holding other factors constant. This should be interpreted as a statistical rather than a causal relationship.

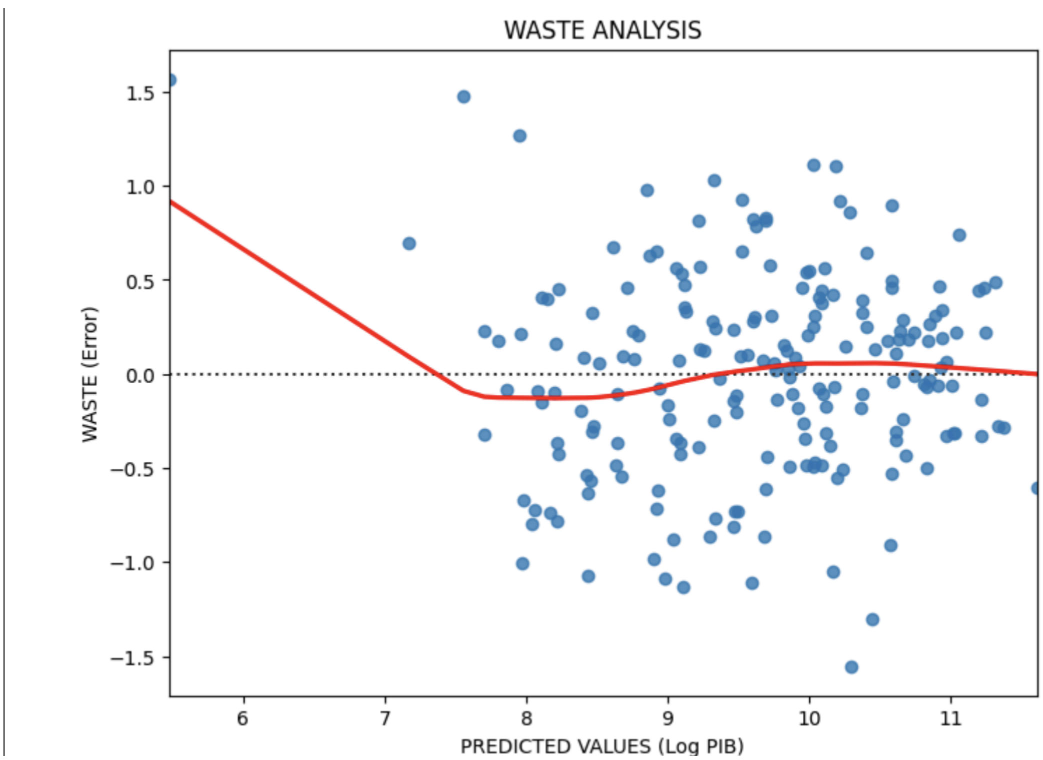
Urban population has a positive and significant coefficient (0.0149), indicating that higher levels of urbanization are associated with higher income per capita.

Life expectancy shows a strong positive effect (0.0937), suggesting that countries with better health conditions and higher human development tend to achieve higher levels of economic prosperity.

The model explains a substantial proportion of the variation in income per capita across countries, with an **R-squared** of 0.778.

All coefficients are statistically significant at the 5% as indicated by their p-values, wich are below 0.05.The t-statistics for all variables are sufficiently large in absolute value, providing strong evidence against the null hypothsis that the coefficients are equal to zero. And the relatively small standar errors indicate that the estimated coefficients are precise.

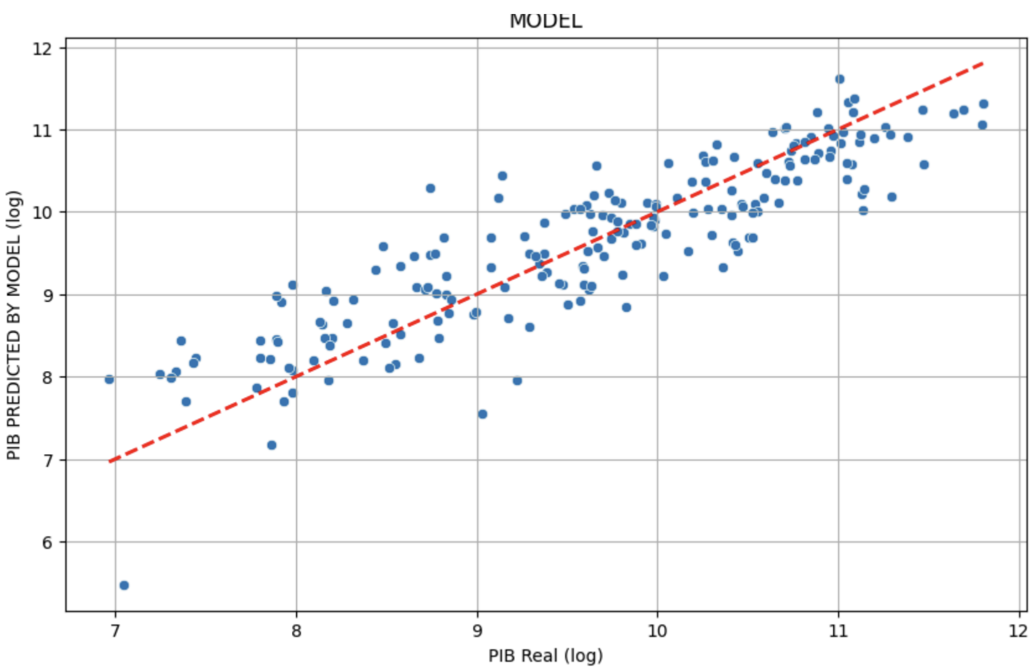
4.1 Model Graphs



The residual analysis suggests that the model assumptions are reasonably satisfied, the residuals appear to be randomly distributed around zero, indicating no clear pattern and supporting the assumption of linearity.

Additionally, there is no strong evidence of heteroscedasticity, as the variance of the residuals appears relatively constant. The distribution of residuals is approximately symmetric, suggesting that the normality assumption is not severely violated.

Overall, the diagnostic plots support the adequacy of the linear regression model.



And now in this graph,to evaluate the predictive performance of the model, a comparison between actual and predicted values is presented.

The results show that the predicted values closely follow the actual values, with most observations lying near the 45-degree line. This indicates that the model has a good fit and is able to capture a significant portion of the variation in GDP per capita across countries.

Some dispersion is observed, which is expected in cross-sectional data; however, the overall pattern suggests that the model provides reasonably accurate predictions. This confirms the model's ability to explain cross-country differences in income per capita.

5 Discussion

The results suggest that structural and human development factors play a key role in explaining differences in income per capita across countries.

The negative association between population and income per capita may reflect structural challenges such as resource distribution, inequality, or varying stages of economic development across countries.

Overall, the findings align with economic intuition and highlight the importance of development indicators in explaining economic prosperity.

6 Limitations

This study has several limitations. First, the analysis is based on cross-sectional data for a single year, which does not capture changes over time.

Second, the model may suffer from omitted variable bias, as other relevant factors such as education, institutional quality, or investment levels are not included.

Finally, the results should be interpreted as associations rather than causal relationships.

7 Conclusion

In conclusion, this study provides evidence that urbanization and life expectancy are positively associated with income per capita, while population size shows a negative relationship within the sample.

These findings highlight the importance of structural and human development factors in explaining differences in economic prosperity across countries. In particular, the strong relationship between life expectancy and income suggests that improvements in health and living conditions are closely linked to higher standards of living.

The model demonstrates a good fit and is able to explain a substantial proportion of the variation in GDP per capita. However, the results should be interpreted with caution, as they reflect statistical rather than causal relationships.

Future research could extend this analysis by incorporating additional variables, such as education or institutional quality, or by using panel data to capture changes over time.

8 References

1. Our World in Data. (2024). Our World in Data. https://ourworldindata.org/
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3. World Bank. (2024). World Development Indicators. https://data.worldbank.org/
4. Montgomery, D. C., Peck, E. A., Vining, G. G. (2012). Introduction to linear regression analysis (5th ed.). Wiley.